The Perkins stopping time

Let

$$\overline{W}_t = \sup_{s\le t} W_s, \qquad \underline{W}_t = \inf_{s\le t} W_s.$$

Define the lower Perkins boundary by

$$K^*(B) \in \arg\max_{K<B} \frac{C_\mu(B) - C_\mu(K) - K}{B-K}, \qquad B>0,$$

and the upper Perkins boundary by

$$K_*(B) \in \arg\min_{K>B} \frac{C_\mu(B) + B - C_\mu(K)}{K-B}, \qquad B<0.$$

The full Perkins stopping time is then given by

$$\tau^P = \inf\big\{t>0 : W_t \le K^*(\overline{W}_t) \text{ or } W_t \ge K_*(\underline{W}_t)\big\}.$$

Equivalently,

$$\tau^P = \tau_- \wedge \tau_+,$$

where

$$\tau_- = \inf\big\{t>0 : W_t \le K^*(\overline{W}_t)\big\}, \qquad \tau_+ = \inf\big\{t>0 : W_t \ge K_*(\underline{W}_t)\big\}.$$

Thus $K^*(B)$ is the lower stopping boundary associated with a running maximum equal to B, while $K_*(B)$ is the upper stopping boundary associated with a running minimum equal to B.